UCS3461 Foundations of Artificial Intelligence

PROBABILITY THEORY

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Review

Probabilistic Inference

- Observed variables (evidence, data): Agent observes/knows certain things about the state of the world (sensor data or symptoms or causes)
- Unobserved variables (hypothesis): Agent needs to reason about other unobserved aspects (where is an object? what is the disease? what is the effect?)
- Model: Agent knows something about how the observed variables relate to the unobserved variables.
- Probabilistic inference/reasoning: a framework for reasoning with uncertain knowledge/belief.

Probability

- Probability is a measure of belief – Bayesian probability or subjective probability.
- Probability theory tells us how to update beliefs based on new information.
- Before an agent observes anything, its belief is its prior probability.
- As it discovers information – typically by observing the environment – it updates its beliefs, giving a posterior probability.
- Reasoning with uncertainty.

Probability: Numerical Measure of Belief/Uncertainty

- Belief in proposition, f, can be measured in terms of a number between 0 and 1 – this is the probability of f.
  - The probability f is 0 means that f is believed to be definitely false.
  - The probability f is 1 means that f is believed to be definitely true.
- f has a probability between 0 and 1, means the agent is ignorant of its truth value.
- Probability is a measure of an agent's belief/ignorance.
- Probability is not a measure of degree of truth.

Random Variables

- Ω is the sample space – a set of possible worlds (outcomes).
- A random variable is a function from possible outcomes to a range of values.
  - countable (or finite) range – discrete random variable.
  - $\{$true, false$\}$ as the range – Boolean random variable.
  - real numbers as the range – continuous random variable.
- A random variable (RV) models an aspect of the world about which we have an uncertainty
  - $R =$ It will rain today (true, false)
  - $W =$ Weather (sunny, cloudy, hot)
  - $T =$ Time taken to drive from home to work (real number)
- Sometimes, the range of a random variable is called its domain

Probability Distribution

- The proposition that "the state of a random variable X is value x" has a probability is written as $P(X = x)$.
  $0 \leq P(X = x) \leq 1$
- Random variables have probability distributions.
  $\sum_x P(X = x) = 1$

T	P(W)
hot	0.5
cold	0.5
	1

W	P(T)
sun	0.6
rain	0.1
fog	0.3
meteor	0
	1

- $P(X = x) = P(x)$
  $P(W = rain) = P(rain) = 0.1$

Joint Distribution

- Distribution over a set of random variables $X_1, X_2, \ldots, X_n$
- $P(x_1, x_2, \ldots, x_n) = P(X_1 = x_1, X_2 = x_2, \ldots, X_n = x_n)$
- $\sum_{x_1, x_2, \ldots, x_n} P(X_1, X_2, \ldots, X_n) = 1$
- P(T, W)

T	W	P(T,W)
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3
		1

- Size of distribution if n variables with domain sizes d?
  $\underbrace{d \times d \times \ldots \times d}_{n \text{ times}} = d^n$
- For all but the smallest distributions, impractical to write out!

Probability Model

- A probability model is a joint distribution over a set of random variables
- Probability model:
  - Random variables, with domains
  - A joint distribution describes how likely each RV assignment of values.

Events

- An event E is a set of outcomes

$$P(E) = \sum_{(x_1, x_2, \ldots, x_n) \in E} P(X_1, X_2, \ldots, X_n)$$

- From a joint distribution, we can calculate the probability of any event
  - Probability that it's hot?
  - Probability that it's hot AND sunny?
  - Probability that it's hot OR sunny?
- Typically, events are partial assignments, like $P(T = hot)$

Marginal Distribution

- Marginal distributions are sub-tables with unneeded variables eliminated.
- Marginalize (sum out): Combine collapsed rows by adding.

$$P(X_1 = x_1) = \sum_{X_2} P(X_1 = x_1, X_2)$$

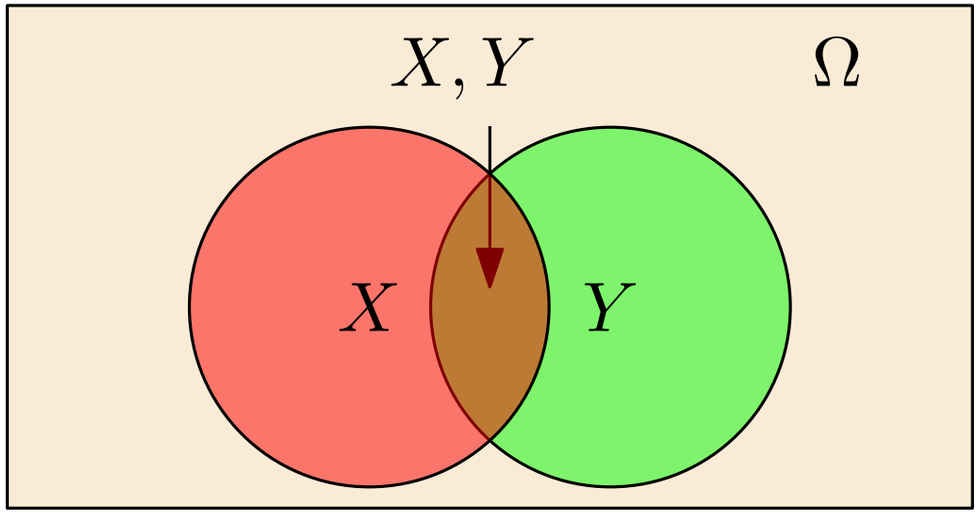
T	W	P(T, W)
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

T	P(T)
hot	0.5
cold	0.5

W	P(W)
sun	0.6
rain	0.4

Conditional Probability

$$P(X|Y) = \frac{P(X,Y)}{P(Y)}$$



T	W	P(T, W)
hot	sun	0.4
hot	rain	0.1
cold	sun	0.2
cold	rain	0.3

$$P(W = s|T = c) = \frac{P(W = s, T = c)}{P(T = c)}$$

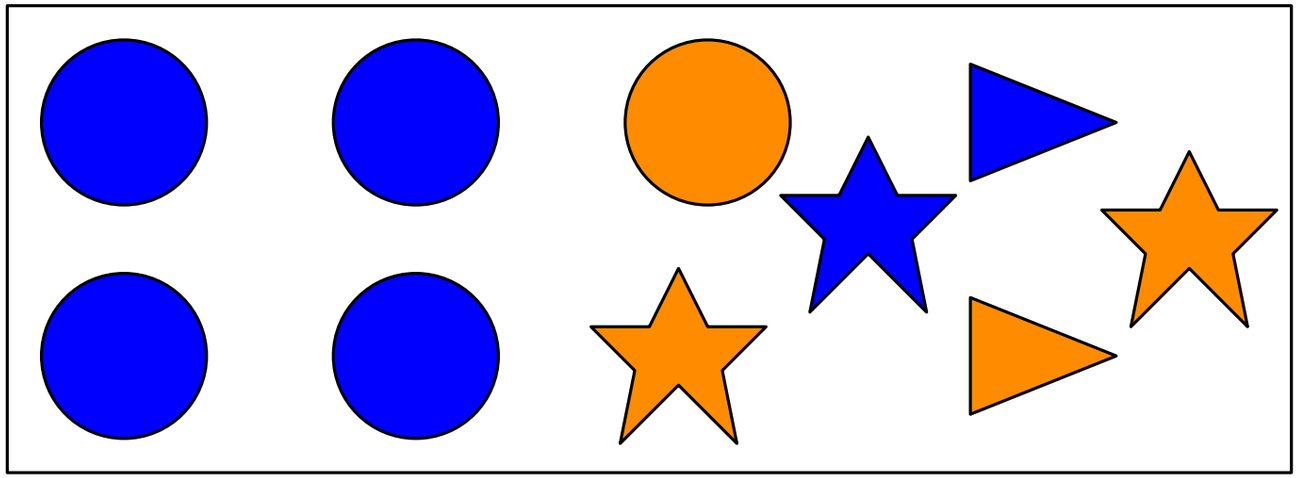
$$= \frac{P(T = c, W = s)}{\sum_w P(T = c, W = w)}$$

$$= \frac{P(T = c, W = s)}{P(T = c, W = s) + P(T = c, W = r)}$$

$$= \frac{0.2}{0.2 + 0.3} = \frac{0.2}{0.5} = 0.4$$

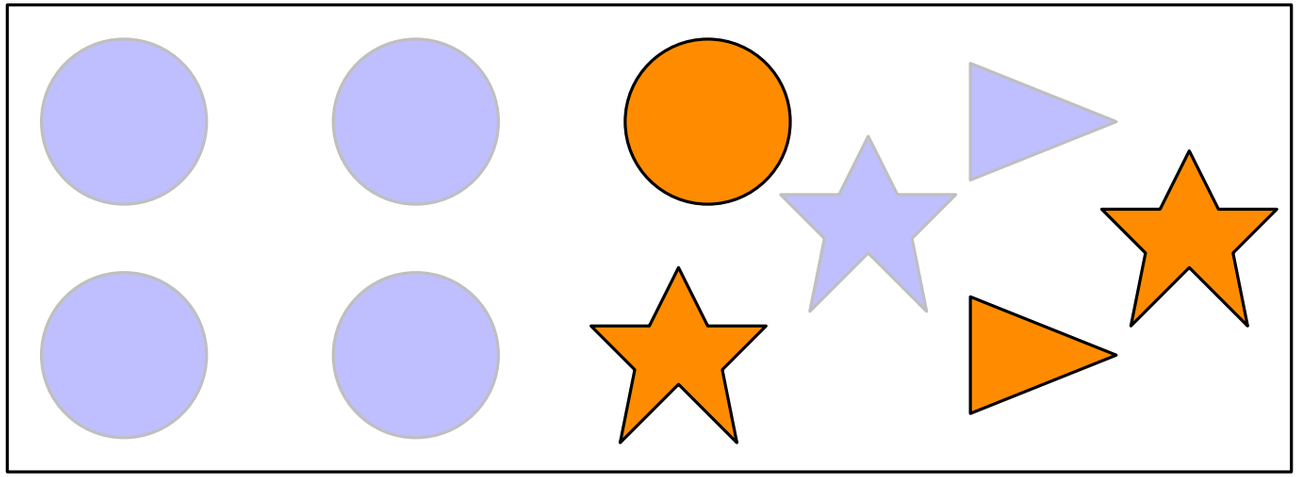
Conditional Probability Distribution

- Probability distribution over some variables, given fixed values of others.



$P(\text{Shape} = \text{star}) = 0.3$
$P(\text{Shape} = \text{circle}) = 0.5$
$P(\text{Shape} = \text{triangle}) = 0.2$

Observe Color = orange



$P(\text{Shape} = \text{star} \mid \text{Color} = \text{orange}) = 0.5$
$P(\text{Shape} = \text{circle} \mid \text{Color} = \text{orange}) = 0.25$
$P(\text{Shape} = \text{triangle} \mid \text{Color} = \text{orange}) = 0.25$

Sum Rule, Product Rule, Bayes Rule

Sum Rule, Product Rule, Bayes Rule

- Sum rule

$$P(X) = \sum_Y P(X, Y)$$

- Product rule

$$P(X, Y) = P(X)P(Y|X) = P(X|Y)P(Y)$$

$$P(X|Y) = \frac{P(X, Y)}{P(Y)}$$

- Bayes rule

$$P(X|Y) = \frac{P(Y|X)P(X)}{P(Y)}$$

Sum Rule (Marginalization)

$$P(X) = \sum_{Y} P(X, Y)$$

$$P(X) = \sum_{Y} \sum_{Z} P(X, Y, Z)$$

$$P(X) = \sum_{Y,Z} P(X, Y, Z)$$

$$P(X_1, \ldots, X_{i-1}, X_{i+1}, \ldots, X_n) = \sum_{X_i} P(X_1, X_2, \ldots, X_{i-1}, X_i, X_{i+1}, \ldots, X_n)$$

Marginalization: Example

P(B,L)

	$B = c$	$B = e$	$B = m$	$P(L) = \sum_B P(B, L)$
$L = j$	0.35	0.10	0.05	0.50
$L = p$	0.28	0.12	0.00	0.40
$L = a$	0.06	0.00	0.04	0.10
$P(B) = \sum_L P(B, L)$	0.69	0.22	0.09	1

Product Rule (Chain Rule)

- Conditional probability $P(Y|X) = \dfrac{P(X,Y)}{P(X)}$
- Product rule

$$P(X,Y) = P(X)P(Y|X)$$
$$P(X_1,X_2,X_3) = P(X_1|X_2,X_3)P(X_2|X_3)P(X_3)$$
$$P(X_1,X_2,X_3,X_4) = P(X_1|X_2,X_3,X_4)P(X_2|X_3,X_4)P(X_3|X_4)P(X_4)$$
$$P(X_1,X_2,X_3,\ldots,X_n) = P(X_n)\prod_{i=1}^{n-1} P(X_i|X_{i+1},X_{i+2},\ldots,X_n)$$

Product Rule: Example

P(L)

L	P(L)
L = j	0.5
L = p	0.4
L = m	0.1
	1

P(B|L)

P(B|L)

L|B	B = c	B = e	B = m	$\sum_B$P(B|L)
L = j	0.7	0.2	0.1	1
L = p	0.7	0.3	0.0	1
L = a	0.6	0.0	0.4	1

P(B,L) = P(B|L) × P(L)

L|B	B = c	B = e	B = m	P(L)
L = j	0.35	0.10	0.05	0.50
L = p	0.28	0.12	0.00	0.40
L = a	0.06	0.00	0.04	0.10
P(B)	0.69	0.22	0.09	1

Bayes Rule

$$P(Y|X) = \frac{P(X|Y)P(Y)}{P(X)}$$

$$P(H|D) = \frac{P(D|H)P(H)}{P(D)}$$

$$P(\theta|D) = \frac{P(D|\theta)P(\theta)}{P(D)}$$

From Causal to Diagnostic

- From causal direction to diagnostic direction

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

$$P(\text{disease}|\text{symptom}) = \frac{P(\text{symptom}|\text{disease})P(\text{disease})}{P(\text{symptom})}$$

Conditional Independencies

Independence of Random Variables

- $P(X, Y) = P(X)P(Y|X)$
- X and Y are independent.

$$P(X, Y) = P(X)P(Y)$$
$$P(Y|X) = P(X)$$

Conditional Independence of Random Variables

X and Y are conditionally independent on Z

$$P(X, Y|Z) = P(X|Y, Z)P(Y|Z) = P(X|Z)P(Y|Z)$$
$$P(X|Y, Z) = P(X|Z)$$
$$P(Y|X, Z) = P(Y|Z)$$

Model and Inference

- Model by a probability distribution $P(X_1, X_2, \ldots, X_n)$
  - Random variables
  - Conditional independencies
  - Conditional Probability Tables (CPT)
- From $P(X_1, X_2, \ldots, X_n)$ derive $P(X_i)$ or $P(X_i|X_j, \ldots, X_k)$

Conditional Independence: Example

One morning Anbu leaves his house and realises that his grass is wet. Is it due to overnight rain or did he forget to turn off the sprinkler last night? Next he notices that the grass of his neighbour, Babu, is also wet. This explains away to some extent the possibility that his sprinkler was left on, and he concludes therefore that it has probably been raining.

Conditional Independence: Example

R	P(R)
0	0.8
1	0.2

S	P(S)
0	0.9
1	0.1

B	R	P(B|R)
0	0	0.8
0	1	0
1	0	0.2
1	1	1

A	R	S	P(A|R,S)
0	0	0	1
0	0	1	0.1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0.9
1	1	0	1
1	1	1	1

Conditional Independence: Example

$P(R = 1)$
0.2

$P(S = 1)$
0.1

R	$P(B = 1 \mid R)$
0	0.2
1	1

R	S	$P(A = 1 \mid R,S)$
0	0	0
0	1	0.9
1	0	1
1	1	1

Conditional Independence

- Without conditional independence
  - $P(A, B, R, S) = P(A|B, R, S)P(B|R, S)P(R|S)P(S)$
  - $2^3 + 2^2 + 2^1 + 2^0 = 8 + 4 + 2 + 1 = 15$
- With conditional independence
  - $P(A, B, R, S) = P(A|R, S)P(B|R)P(R)P(S)$
  - $2^2 + 2^1 + 1 + 1 = 8$

Inference

$$
\begin{aligned}
&P(S=1|A=1)\\
&= \frac{P(S=1, A=1)}{P(A=1)}\\
&= \frac{\sum_{B,R} P(A=1, B, R, S=1)}{\sum_{B,R,S} P(A=1, B, R, S)}\\
&= \frac{\sum_{B,R} P(A=1|R, S=1)P(B|R)P(R)P(S=1)}{\sum_{B,R,S} P(A=1|R, S)P(B|R)P(R)P(S)}\\
&= \frac{\sum_{R} P(A=1|R, S=1)P(R)P(S=1)}{\sum_{R,S} P(A=1|R, S)P(R)P(S)}\\
&= \frac{0.9 \times 0.8 \times 0.1 + 1 \times 0.2 \times 0.1}{0.9 \times 0.8 \times 0.1 + 1 \times 0.2 \times 0.1 + 0 \times 0.8 \times 0.9 + 1 \times 0.2 \times 0.9}\\
&= 0.3382
\end{aligned}
$$

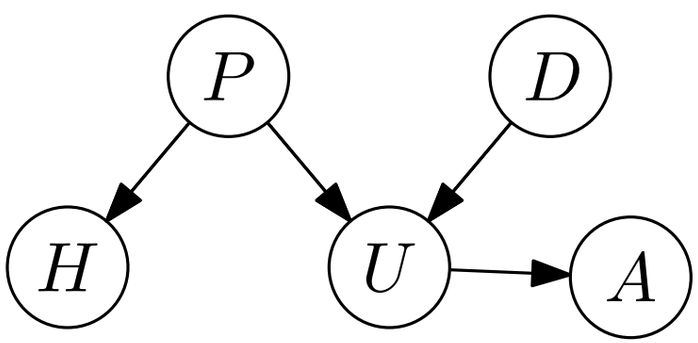
B	R	P(B|R)
0	0	0.8
0	1	0
1	0	0.2
1	1	1

A	R	S	P(A|R,S)
0	0	0	1
0	0	1	0.1
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0.9
1	1	0	1
1	1	1	1

Exercise 1

(Party Animal). The party animal problem corresponds to the network. The boss is angry and the worker has a headache – what is the probability the worker has been to a party? To complete the specifications, the probabilities are given as follows:

$$p(U=1|P=1,D=1)=0.999 \qquad p(U=1|P=0,D=1)=0.9$$
$$p(U=1|P=1,D=0)=0.9 \qquad p(U=1|P=0,D=0)=0.01$$
$$p(H=1|P=1)=0.9 \qquad p(H=1|P=0)=0.2$$
$$p(A=1|U=1)=0.95 \qquad p(A=1|U=0)=0.5$$
$$p(P=1)=0.2 \qquad p(D=1)=0.4$$



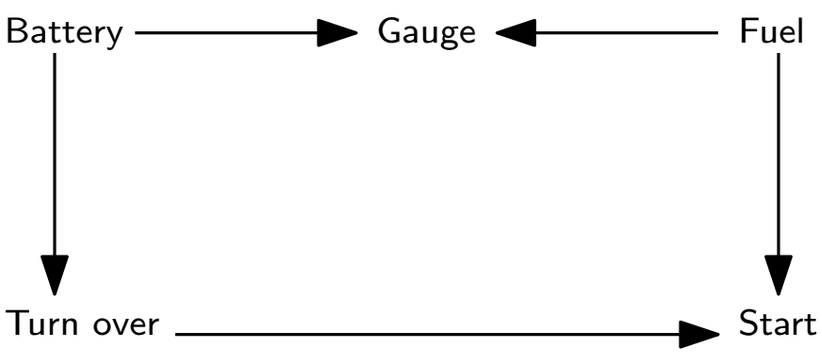
- D = Demotivated at work
- U = Underperform at work
- P = Been to Party
- H = Got a Headache
- A =Boss Angry

Exercise 2

The network concerns the probability of a car starting, with

$$p(g = \text{empty} | b = \text{good}, f = \text{not empty}) = 0.04$$
$$p(g = \text{empty} | b = \text{good}, f = \text{empty}) = 0.97$$
$$p(g = \text{empty} | b = \text{bad}, f = \text{empty}) = 0.99$$
$$p(g = \text{empty} | b = \text{bad}, f = \text{not empty}) = 0.1$$
$$p(s = \text{fa} | t = \text{tr}, f = \text{not empty}) = 0.01$$
$$p(s = \text{fa} | t = \text{fa}, f = \text{empty}) = 0.99$$
$$p(s = \text{fa} | t = \text{tr}, f = \text{empty}) = 0.92$$
$$p(s = \text{fa} | t = \text{fa}, f = \text{not empty}) = 1.0$$
$$p(t = \text{fa} | b = \text{good}) = 0.03$$
$$p(t = \text{fa} | b = \text{bad}) = 0.98$$
$$p(f = \text{empty}) = 0.05$$
$$p(b = \text{bad}) = 0.02$$

Calculate $P(f = \text{empty} | s = \text{no})$, the probability of the fuel tank being empty conditioned on the observation that the car does not start.



Summary

- Rules of probability: sum rule and product rule
- Bayes theorem
- Inference using Bayes theorem

End